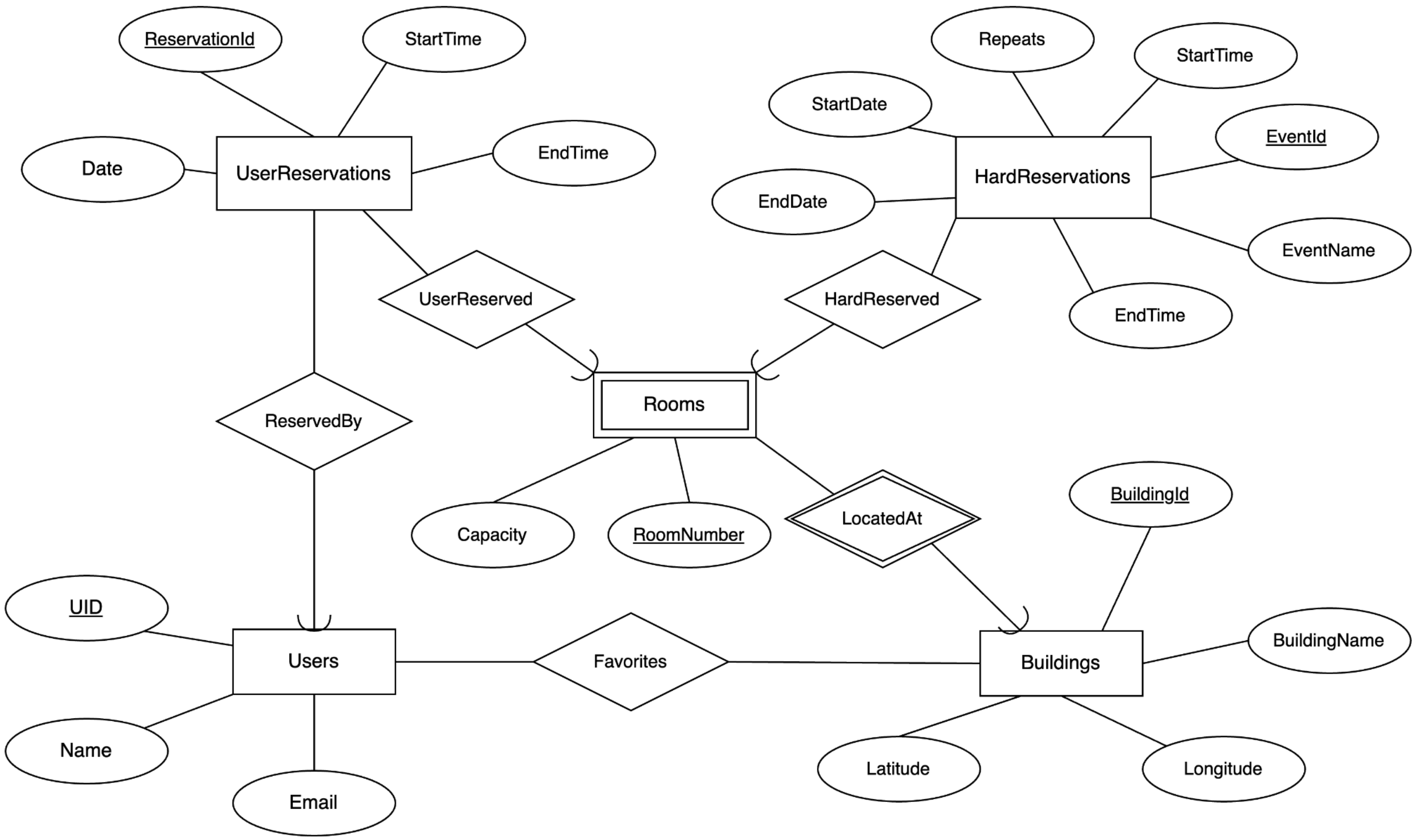
IlliniSpaces ER Diagram and Relational Schema



Entities

- Users: modeled as an entity because it represents the individual users who have multiple attributes.
    - Each User can have **zero or more** UserReservations, as long as they do not overlap in terms of calendar time.
    - Each User can have **zero or more** “favorited” Buildings.
- Buildings: modeled as an entity because it represents the physical buildings on campus. It is not an attribute for Rooms because it has its own attributes.
    - Each Building can be “favorited” by **zero or more** Users.
    - Each Building can be the location for **one or more** Rooms, because each building can have multiple rooms, but it should have at least one room.
- Rooms: modeled as a weak entity because rooms are uniquely identifiable with the building id, and it represents the physical rooms on campus.

    - Each Room can only be located at **exactly one** Building. We assume there is no room that simultaneously belongs to two or more Buildings.
    - Each Room can be HardReserved by **zero or more** HardReservations (classes held by university), because each room can host different classes throughout the week.
    - Each Room can be UserReserved by **zero or more** UserReservations (unofficial reservations by the app users), because each room can be reversed by different users throughout a day as long as the reservations do not overlap.
- HardReservations: modeled as an entity because it is the (sections of) individual courses held by the university.
    - Each HardReservation can HardReserve **exactly one** room, because each hard reservation represents an officially scheduled university event or class that occurs in a specific Room at a defined time.
    - Each HardReservation can HardReserve exactly one Room because university events are scheduled at a specific location, but each Room can be HardReserved by **zero or more** HardReservations throughout the semester.
    - Each HardReservation has attributes including StartDate, EndDate, Repeats, StartTime, EndTime, and EventName, which define when and how often the event occurs.
    - Each HardReservation has an EventId, which uniquely identifies it.
- UserReservations:
    - Each UserReservation can UserReserve **exactly one** room, because each user reservation represents a specific booking made by an app user to use a Room at a defined time.
    - Each UserReservation can UserReserve exactly one room because a reservation applies to a single space at a specific time, but each Room can be UserReserved by **zero or more** UserReservations as long as the reservations do not overlap.
    - Each UserReservation has attributes including Date, StartTime, and EndTime, which specify the timeframe for the reservation.
    - Each UserReservation has a ReservationId, which uniquely identifies it.

Relational Schema:

1. UserReservations(ReservationId:INT [PK], Date: VARCHAR(10), StartTime:VARCHAR(10), EndTime:VARCHAR(10), UID:INT [FK to Users.UID], RoomNumber:INT [FK to Rooms.RoomNumber], BuildingId:INT [FK to Rooms.BuildingId])

<u>FD’s:</u>

- ReservationId -> all attributes

2. HardReservations(EventId:INT [PK], StartDate: VARCHAR(10), EndDate: VARCHAR(10), Repeats: VARCHAR(10), StartTime:VARCHAR(10), EndTime:VARCHAR(10), EventName:VARCHAR(100), RoomNumber:INT [FK to Rooms.RoomNumber], BuildingId:INT [FK to Rooms.BuildingId])

<u>FD’s:</u>

- EventId -> all attributes

3. Rooms(RoomNumber:INT [PK], BuildingId:INT [PK] [FK to Buildings.BuildingId], Capacity:INT)

<u>No non-trivial FD’s.</u>

4. Users(UID:INT [PK], Email:VARCHAR(50), FullName:VARCHAR(50))

<u>FD’s:</u>

- UID -> all attributes
- Email -> all attributes

5. Buildings(BuildingId:INT [PK], BuildingName:VARCHAR(75), Longitude:DECIMAL(11, 8), Latitude:DECIMAL(10, 8))

<u>FD’s:</u>

- BuildingId -> all attributes
- Longitude, Latitude -> all attributes

6. Favorites(UID:INT [PK] [FK to Users.UID], BuildingId:INT [PK] [FK to Buildings.BuildingId])

<u>No non-trivial FD’s.</u>

These relations are in BCNF since for all FD’s, the left hand side is a superkey of the relation.

Assumptions:

- We assume two distinct buildings can have the same name though it’s unlikely in practice. Thus, BuildingName does not determine Longitude, Latitude.
- Cascade deletes (RHS has a foreign key to a LHS entry):
  - Users -> User Reservations
  - Rooms -> User and Hard Reservations
  - Buildings -> Rooms
- Room + Time interval should be unique across both HardReservations and UserReservations, and no intervals should intersect for the same room. We will enforce this using triggers/constraints.
- NOT NULL should be applied to all attributes (except HardReservations.Repeats), especially the many to (exactly) 1 FK's (ER diagram semicircles).